



World First 9v Battery That Can Be Recharged From a USB Port

By Imetomi in Circuits > Electronics 31,802 198 11 ★ Featured



The 9v batteries are very often used in electronics, example: programmable circuits, robots and etc... But they have a very huge problem, theirs capacity is low and the rechargeable 9v batteries needs at least 9.5v to charge them up. I thought much time about this project because would be an important and useful power supply for many project. This 9v battery allows you to charge it up from a USB port in 22 minutes. It has the same capacity (or higher) that an average 9v battery so the usage time is the same. You can use in any household devices like a scale or you can power up a 7805 regulator for a phone charging. It can drive even a CPU fan. There is inside a 3.7v Lithium Ion battery with 200mAh capacity. But can be powered from a 3v coin cell battery. Fits perfectly in an old plastic 9v battery case. See on my multimeter that the voltage is 8.95v. At this voltage the current is much

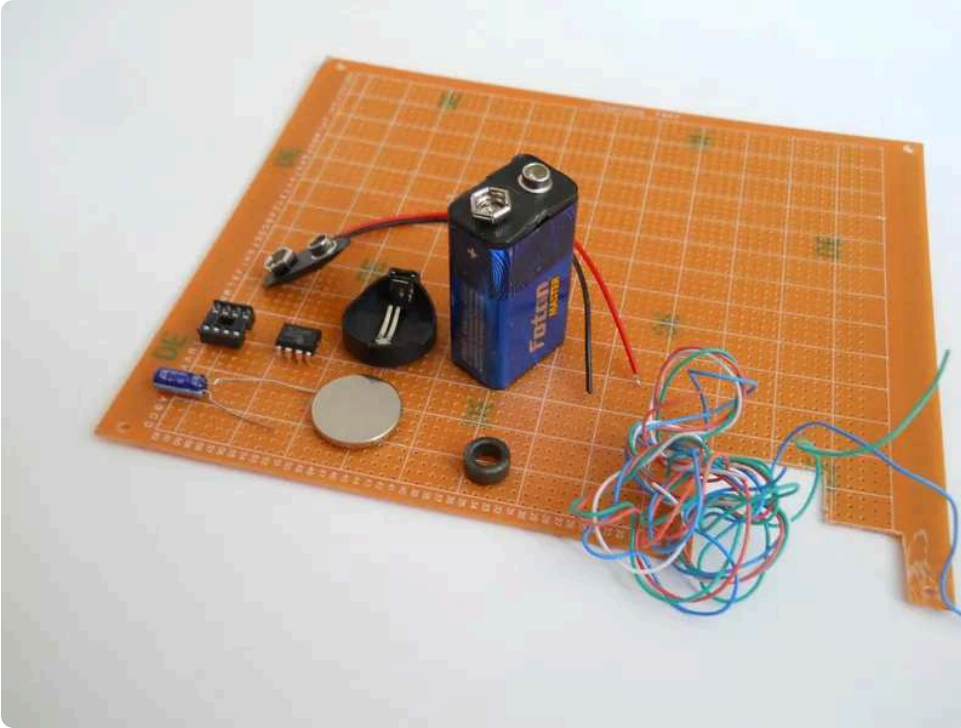
higher than a simple 9v battery. The base of the circuit is the TL496 CP integrated circuit. This IC can boost low voltage to 9 volts with the help of a 47uF cap and an inductor.

With this new invention you can save lots of money and never need to buy anymore 9v batteries. Simply plug into your laptop, desktop PC or wallpug and charge it up in 20 minutes.

Now go gather parts!



Step 1: Gather Tools and Materials



You need to know that I can't find the best and the cheaper links for the parts. Here in my country the total cost was less than 5 bucks, but in the US the electronic components are very expensive. I tried to find the best links for the components. But on the ebay the TL496 was \$2 and here only \$0.1. Sorry :(.

Tools:

- soldering iron
- cutting tools
- glue gun
- and a multimeter

Parts:



First choose your power supply! This would be a 3.7v 200mAh Li-ion battery or a 3v coin cell battery. I recommend the Li-Ion, it's much more effective.

You'll need an old 9v battery that has a plastic case.

9v battery clip from your batttery

47uF capacitor

8 pin IC socket

TL496CP 9v step up IC

toroidal inductor

breadboard (optional for ptototyping)

PCB board

wires

1k resistor

a diode and a low power LED (red, or green)

switch (optional but I don't used)

Some advices:

The Li-ion batteries are much more effective, but I can't put my charging circuit in the 9v battery case. There's no enough space so I calculated the charging time. The formula is: $\text{hours} = \text{capacity (in mAh)} / \text{current (in mA)}$. In my case this is $\text{hr} = 200 / 500$ so a 200mAh battery from a USB port charges up in 0.4 hours. This is 24 minutes.

Never leave the battery in the charger for more than this calculated time!

It's DANGEROUS!

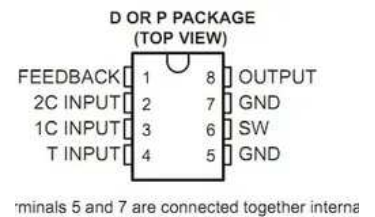
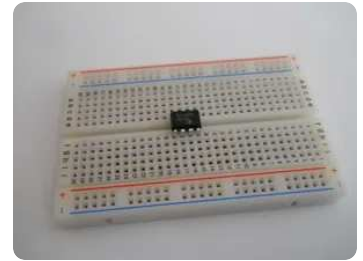
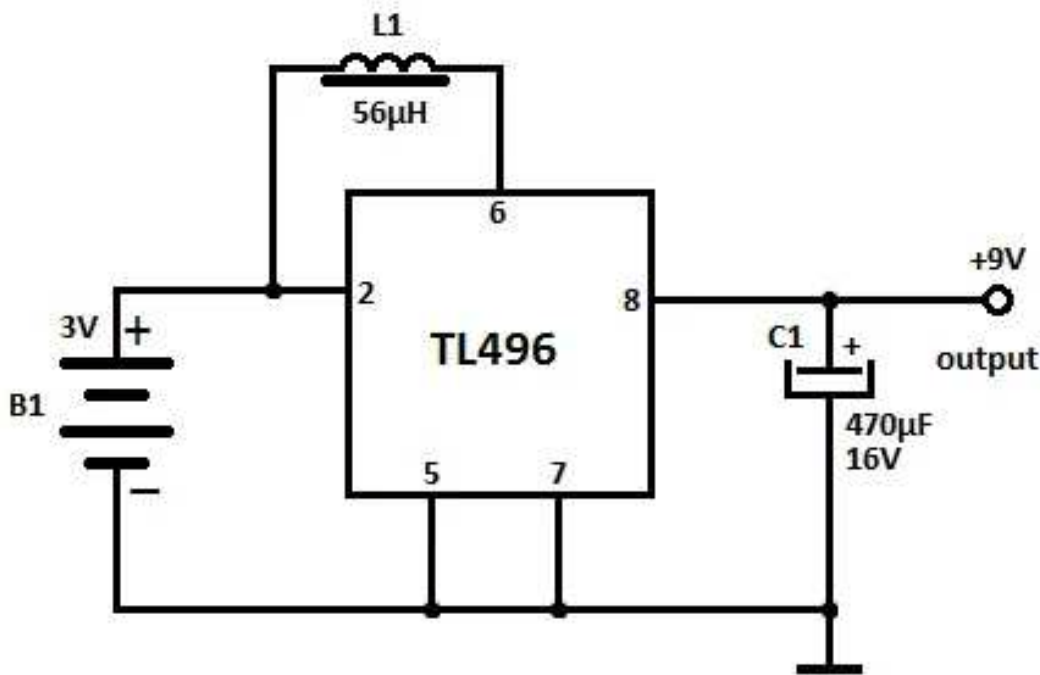
The maximum output voltage would be 9.2 volts. Depends on the capacitor value. Higher uF capacitor gives higher voltage. But works perfect in this range: 5-470uF. I used a 47uF because no requires much space in the 9v battery box.

If you're ready go make a prototype!





Step 2: Prototyping



See the circuit on the first image, I replaced the 3v battery to my 200mAh 3.7v battery. Works much better!

First install the IC on your breadboard.

Connect the 5 and 7 pins together with a jumper wire.

Then make your inductor with a 24# awg wire. I made 11 turns.

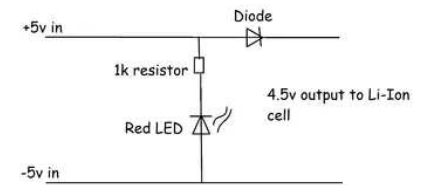
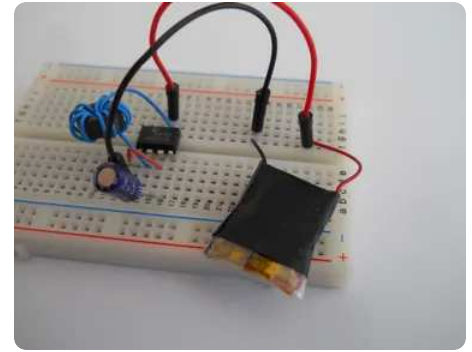
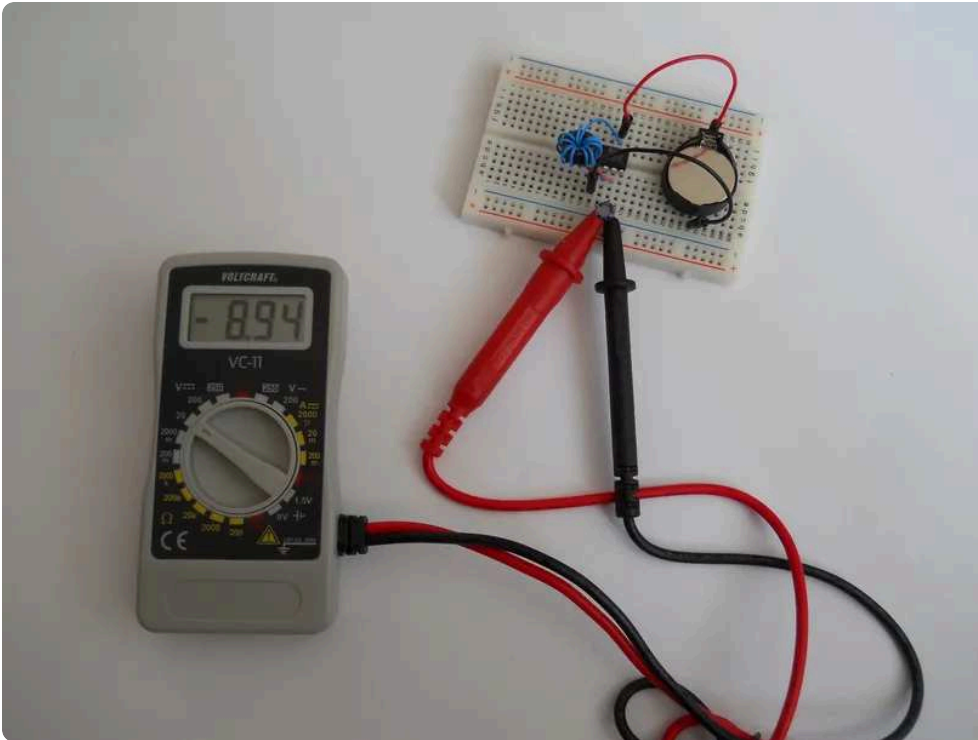
Connect the inductor to your breadboard, to pins 2 and 6.

Put the capacitor to the output of the IC, pins 7 and 8.

If you made these steps install your power supply with jumper wires. First I tried with a 3v lithium battery but the amperage was very low so I searched a Li-ion cell.



Step 3: Test the Prototype!

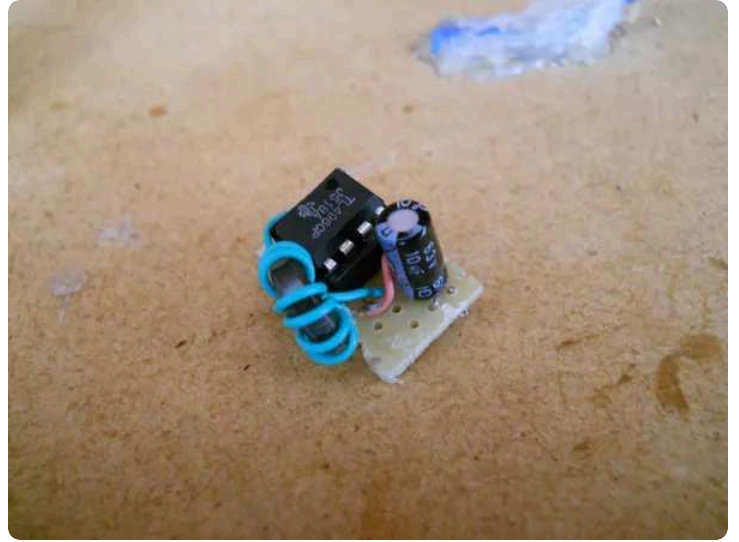


The circuit works perfect, the better version is on the 2nd image. I think no need more explanation. If your circuit works go solder together.

I used a diode, a resistor and an LED for the charger circuit. The LED will light when the battery is charging. **On the circuit the LED mark has reversed polarity, sorry. I'll try to correct it soon.**



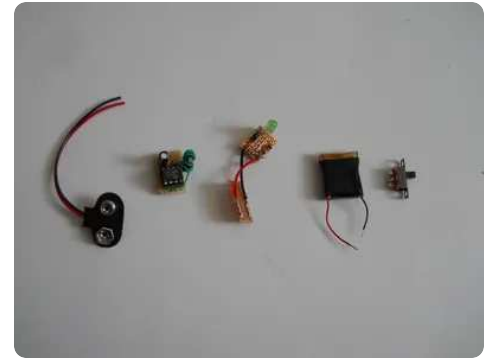
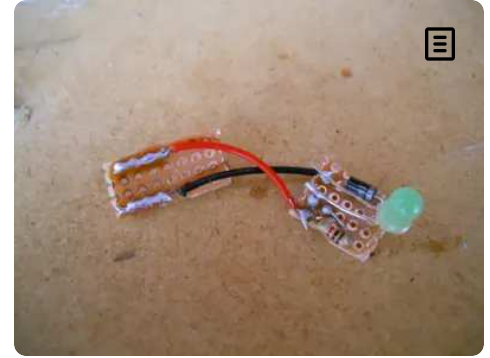
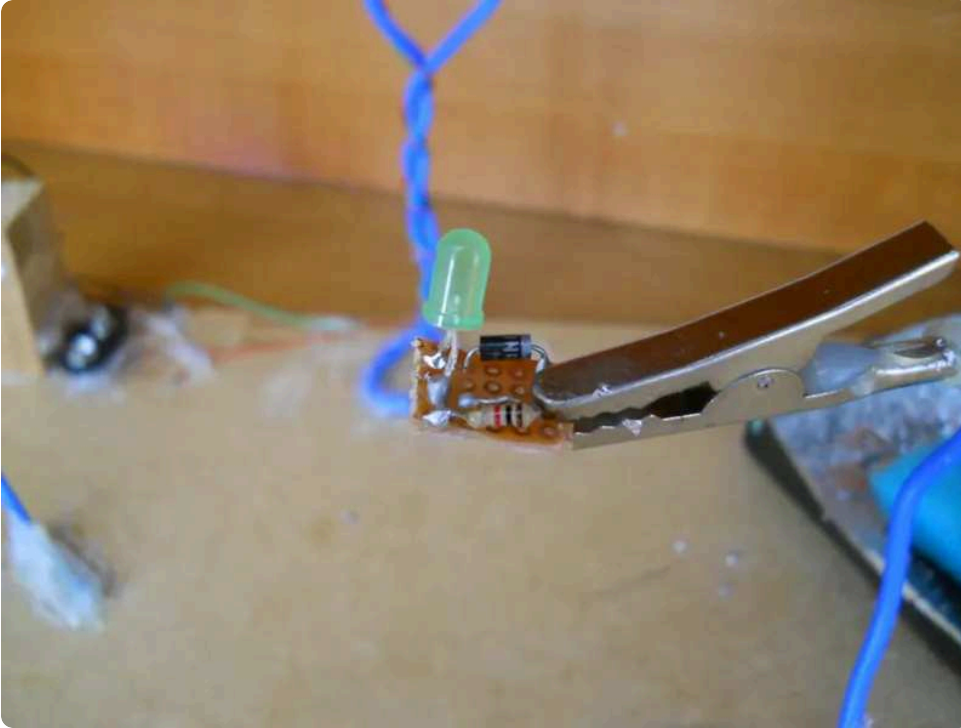
Step 4: Solder the 9v Step Up



I soldered first the 9v step up circuit because this is a little bit difficult. But seeing the connections on the circuit diagram everybody can build easily. Used a socket for the TL496 because can't forbear high temperature. The input will be the 2 and 5 pins and the output pins are 7 and 8 where 8 is positive and 7 is positive. These pins are under the capacitor. On the go I exchanged the 47uF capacitor to a 10uF capacitor. The voltage is the same, but requires smaller place. and fits perfectly into the 9v battery case. Try to make your circuit as small like me, you haven't much space for this gadget!



Step 5: The Charger

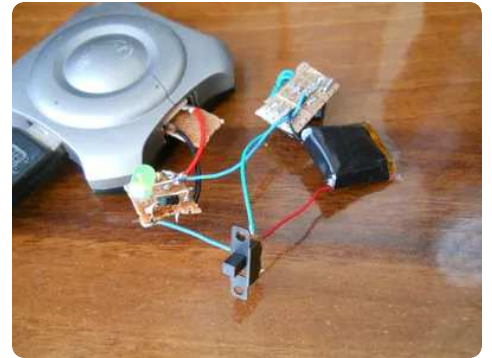
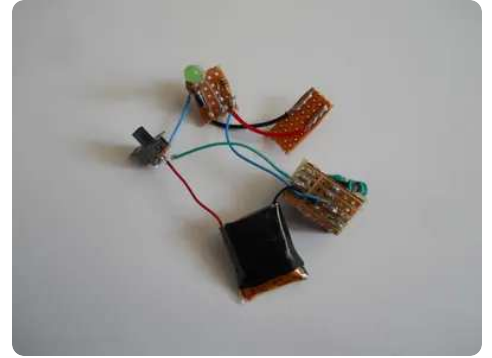
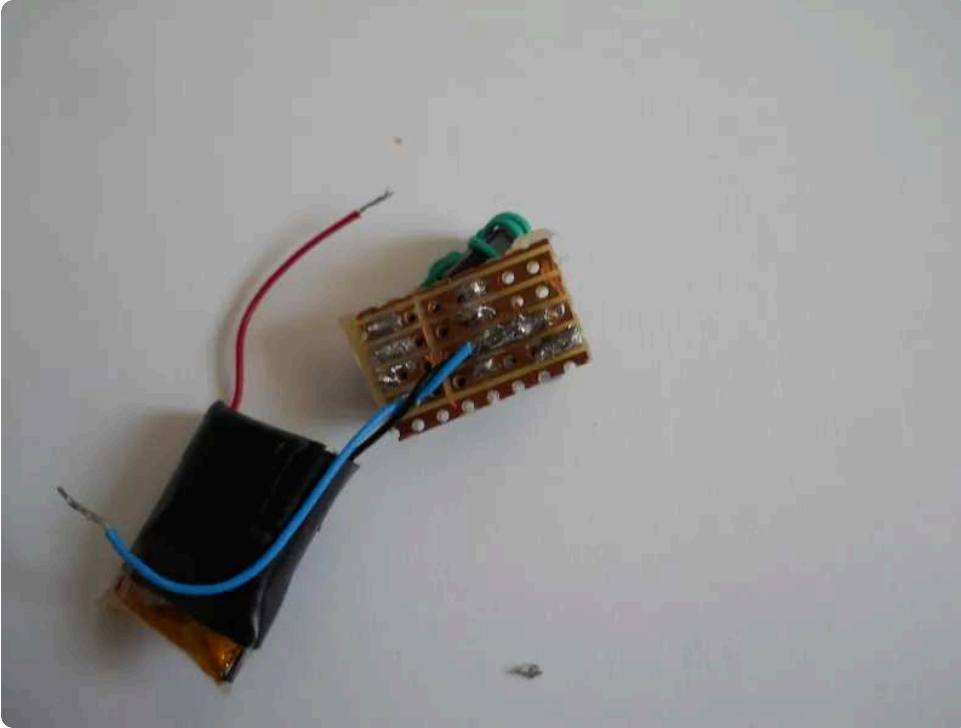


Now make your charger circuit. This is very easy. Simply solder the resistor to the cathode of the LED then the diode to the positive input. This will blocks the current flows backward. I haven't enough space inside the 9v battery case so I couldn't use a regular USB male connector so simply soldered the input wires on another PCB board. Do the same thing you too, just like on the pictures. But if you have more space use a regular male connector. The small LED will light up when the circuit is in the charger. Before you solder together the parts test it everything with your multimeter.

*Finally you'll get a battery, a 9v clip, a charger and a 9v step up circuit. **Solder them together!***



Step 6: Create the Full Circuit



First connect the negative wire of the Li-Ion cell to the 5 or 7 pin of the 9v step up circuit. Then solder an another wire to these pins. This will be soldered to the negative input of the charger. The positive wire of the battery needs to be soldered to the 2nd pin just like on our prototype. To this pin comes the positive input of the charger. See the circuit diagram on the last image.

Hope I explained right and understandable.

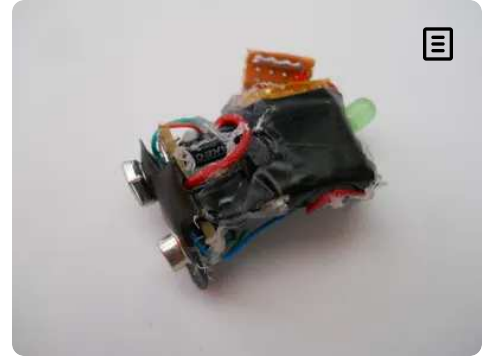
Then you'll get a circuit like on the 2nd image. There's a switch, but later I removed because is meaningless.

If you're done with this test it. If you plug your circuit in a USB port the LED musts light. Then test the output voltage too. If everything works get your glue gun and try to make the circuit as small that fits easily in the 9v battery box.





Step 7: Glue!



Under the 9v clip I put the 9v step up circuit and glued. The battery is glued under this circuit and above the battery is the charger circuit. This is the easiest design and fits perfectly in the 9v battery case.

Now leave the device for at least 20 minutes in the charger then put back in the case.



Step 8: Advantages



Some advantages:

- These Li-Ion cells can be recharged 1000 times so you never need to buy anymore a regular 9v battery.
- The weight of this device is the quarter of a simple 9v battery. It has the same capacity.
- **The 9 volt output CAN'T BE SHORTED!** The IC stops working if you short the circuit and when you make the circuit works again correct the IC



starts working. SO NO NEED TO WORRY ABOUT OVER HEAT.



Step 9: Test and THE END



Install to any household item or DIY project and will work perfectly. Works also with the very popular 7805 regulators, that charges your phone. Can't charge much percents but is good in the case of emergency. And fits easily in your pocket.

Hope you liked! Write your opinion and vote on me.

Extracted URLs from **<https://www.instructables.com/1st-9v-Battery-that-can-be-recharged-from-a-USB-Po/>**

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